Inequalities — problem 7.96

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Prove that, for any distribution of signs + and − in the odd powers of x,

$$x^{2n} \pm x^{2n-1} + x^{2n-2} \pm x^{2n-3} + \cdots + x^4 \pm x^3 + x^2 \pm x + 1 > \frac{1}{2}.$$